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Name _____

April 16, 2011 Sunday

Department _____

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University Physics
Sample Midterm Examination
School of Intensive Instruction in Sciences and Arts, Nanjing University

Physical Constants

Gravitational constant

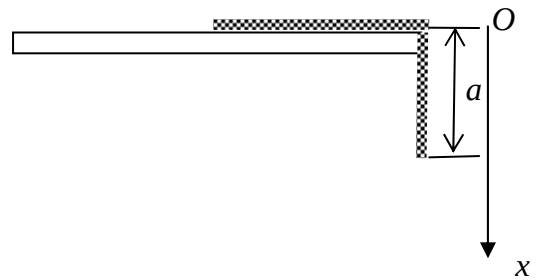
$$G = 6.67 \times 10^{-11} \text{ m}^3 / \text{s}^2 \cdot \text{kg}$$

Standard gravity acceleration

$$g_n = 9.806 \text{ m/s}^2$$

Select five out of following six problems.

(1) (20pts) As shown in the accompanying figure, a uniform chain of total length l hangs with a piece of length a ($0 < a < l$) over the edge of a plane table. The initial velocity of the chain is zero and there are no friction between the surface of the table and the chain.



(a) Find the velocity of the chain at the instant it slides off the table.

(b) Calculate the time in which the chain slides from the table.

Solution: (a) The equation of motion is

$$m\ddot{x} = \frac{m}{l} xg.$$

By integrating both sides with respect to x , and noting that

$$\ddot{x}dx = \frac{d\dot{x}}{dt}dx = \dot{x}\frac{dx}{dt} = \dot{x}\dot{x} = vdv$$

we have

$$\int_0^v vdv = \frac{g}{l} \int_a^x xdx$$

or

$$\frac{1}{2}v^2 = \frac{1}{2}\frac{g}{l}(x^2 - a^2) \quad \text{or} \quad v = \sqrt{\frac{g(x^2 - a^2)}{l}}.$$

At the instant it slides off the table, $x = l$ and

$$v = \sqrt{\frac{9(l^2 - a^2)}{l}}.$$

(b) From the result above, we have

$$\frac{dx}{dt} = \sqrt{\frac{g}{l}} \sqrt{x^2 - a^2}.$$

It can be rewritten as

$$\frac{dx}{\sqrt{x^2 - a^2}} = \sqrt{\frac{g}{l}} dt.$$

Taking integration of both sides of above equation, we obtain

$$\int_a^l \frac{dx}{\sqrt{x^2 - a^2}} = \sqrt{\frac{g}{l}} \int_0^t dt,$$

or

$$t = \sqrt{\frac{l}{g}} \ln(x + \sqrt{x^2 - a^2}) \Big|_a^l = \sqrt{\frac{l}{g}} \ln\left(\frac{l + \sqrt{l^2 - a^2}}{a}\right).$$

(2) (20pts) The potential energy function for the force between two atoms in a diatomic molecule can be express as

$$U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$$

where a and b are positive constants and x is the distance between two atoms.

(a) Find the force between two atoms as the function of x .

(b) What is the equilibrium separation between two atoms?

(c) Find the minimum energy needed to break the molecule apart.

Solution: (a) The force between two atoms is

$$F(x) = -\frac{dU}{dx} = \frac{12a}{x^{13}} - \frac{6b}{x^7}.$$

(b) The equilibrium separation is determined by

$$\left. \frac{dU}{dx} \right|_{x=x_m} = -\frac{12a}{x_m^{13}} + \frac{6b}{x_m^7} = 0.$$

Thus, we have $x_m = \left(\frac{2a}{b}\right)^{1/6}.$

(c) As x approaches ∞ , $U = 0$. Then, the minimum energy needed to break the molecule is

$$E_d = U(\infty) - U(x_m) = -\frac{a}{x_m^{12}} + \frac{b}{x_m^6} = \frac{b^2}{4a}.$$

(3) (20pts) Assume that the Earth's orbit around the Sun is circular, and the Sun's mass is much larger than the Earth's mass.

(a) Compute the ratio of the kinetic and potential energies of the Earth.

(b) The Sun undergoes a supernova explosion and its mass suddenly reduces to half of its original value. Assume that the energy released in the explosion has no effect on the Earth. Compute the ratio of the kinetic and potential energies of the Earth at the instant after the explosion.

(c) What is the new total energy of the Earth? What is the shape of the orbit now? Will the earth escape the solar system?

Solution: (a) For circular motion, we have

$$T = \frac{1}{2} m_e v^2$$

and

$$U = -\frac{GM_s m_e}{R}$$

By equating the gravitational force to the centripetal force, we have

$$\frac{GM_s m_e}{R^2} = m_e \frac{v^2}{R}$$

or

$$\frac{GM_s m_e}{R} = m_e v^2 .$$

So

$$T = \frac{1}{2} m_e v^2 = -\frac{1}{2} U .$$

The ratio is

$$\frac{T}{U} = -\frac{1}{2} .$$

If the mass of the sun suddenly goes to $1/2$ of its original value, the kinetic energy remains unchanged but the potential energy is halved:

$$T' = T, \quad U' = \frac{U}{2} .$$

The ratio then is

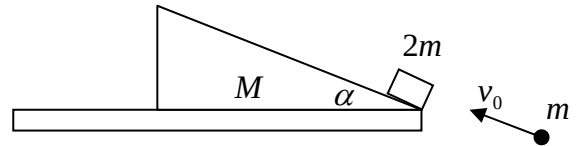
$$\frac{T'}{U'} = -1 .$$

(c) The total energy is

$$E' = T' + U' = 0 .$$

Then, the orbit is a parabola and the earth will escape the solar system.

(4) (20pts) A slider of mass $2m$ is initially at rest at the bottom edge of a wedge of mass M and angle α . The wedge has a frictionless surface and rests on a frictionless table. At $t = 0$, a bullet of mass m and velocity v_0 traveling parallel to the upper surface of the wedge hits and sticks in the slider.



(a) What is the maximum height reached by the slider?

(b) What is the wedge's velocity at the time the slider reaches its maximum height?

(c) How far has the wedge moved at that time?

Solution: (a) Immediately after the bullet hits and sticks in the slider, we have

$$mv_0 = (2m + m)v .$$

Thus, the velocity of the slider (with the bullet) is $v = v_0/3$. Assume the maximum height of the slider is $h_{\max}$, then the conservations of energy gives

$$\frac{1}{2} (3m)v^2 = \frac{1}{2} (M + 3m)V^2 + 3mgh_{\max} . \quad (1)$$

And the conservation of the momentum gives

$$3mv \cos \alpha = (M + 3m)V . \quad (2)$$

Eliminating V in Eq.(1) with the use of Eq.(2), we have

$$\frac{1}{6}mv_0^2 = \frac{1}{2} \frac{m^2 \cos^2 \alpha}{M+3m} v_0^2 + 3mgh_{\max}.$$

Thus, we obtain

$$h_{\max} = \frac{v_0^2}{6g} \left(\frac{1}{3} - \frac{m \cos^2 \alpha}{M+3m} \right). \quad (3)$$

(b) From Eq. (2), we have

$$V = \frac{3mv \cos \alpha}{M+3m} = \frac{mv_0 \cos \alpha}{M+3m}.$$

(c) The equation of motion for the slider (in the wedge frame) can be written as

$$\begin{aligned} -N \sin \alpha - 3m\ddot{X} &= 3m\ddot{x} \\ N \cos \alpha - 3mg &= 3m\ddot{y} \\ dy &= dx \tan \alpha \end{aligned} \quad (4)$$

And the equation of motion for the wedge is

$$M\ddot{X} = N \sin \alpha. \quad (5)$$

The last equation of Eq. (4) gives $\ddot{y} = \ddot{x} \tan \alpha$. Eliminating $\ddot{x}$ and $\ddot{y}$ from Eq.(4), we have

$$-N \sin \alpha - 3m\ddot{X} = (N \cos \alpha - 3mg) \cot \alpha. \quad (6)$$

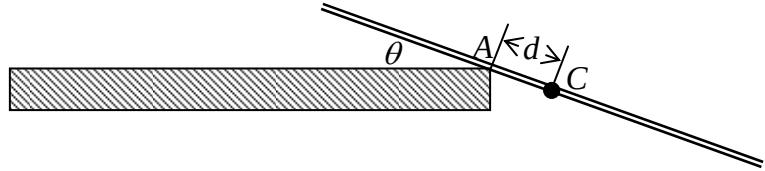
By using Eq.(5) to eliminate N in Eq. (6), we finally have

$$\ddot{X} = \frac{3mg \cot \alpha}{M(1 + \cot^2 \alpha) + 3m} \quad (const.) \quad (7)$$

Thus,

$$t = \frac{V}{\ddot{X}} \quad \text{and} \quad S = \frac{1}{2} \ddot{X} t^2 = \frac{1}{2} \frac{V^2}{\ddot{X}} = \frac{v_0^2}{12g} \frac{m[M(1 + \cot^2 \alpha) + 3m] \sin 2\alpha}{(M+3m)^2}. \quad (8)$$

(5) (20pts) A uniform rod of mass m and length l is held flat on a horizontal table and at right angle to an edge A of the table. The



center of mass C of the rod is beyond the edge of a distance d . The rod is released at rest and starts to rotate about the edge A and eventually slides off the table. The coefficient of static friction between the rod and the edge of the table is μ .

(a) Find the rotational inertias of the rod, I_C about the center of mass C and I_A about point A .

(b) Calculate the angular velocity of the rod as the function of rotation angle θ before sliding occurs.

(c) Determine the angle θ_0 when sliding of the rod begins.

Solution: The rotational inertia about C is

$$I_C = 2 \int_0^{l/2} x^2 \frac{m}{l} dx = \frac{1}{12} ml^2.$$

The rotational inertia about A is

$$I_A = I_C + md^2.$$

(b) The conservation of the mechanical energy gives

$$0 = \frac{1}{2} I_A \omega^2 - mgd \sin \theta.$$

So,

$$\omega^2 = \frac{2mgd \sin \theta}{I_A}, \quad \omega = 2\sqrt{\frac{6gd \sin \theta}{l^2 + 12d^2}}.$$

(3) At $\theta = \theta_0$, when the sliding begins, $F_f = \mu N$. The equation of motion of the center of mass of the rod is then

$$mg \cos \theta_0 - N = md \cdot \ddot{\theta}, \quad (1)$$

$$F_f - mg \sin \theta_0 = m\omega^2 d. \quad (2)$$

The equation of motion for the rotation is

$$mgd \cos \theta_0 = I_A \ddot{\theta}. \quad (3)$$

Eliminating $\ddot{\theta}$ in Eq.(1) by using Eq.(3), we obtain

$$N = mg \cos \theta_0 \left(1 - \frac{md^2}{I_A}\right) = mg \cos \theta_0 \frac{I_C}{I_A}. \quad (4)$$

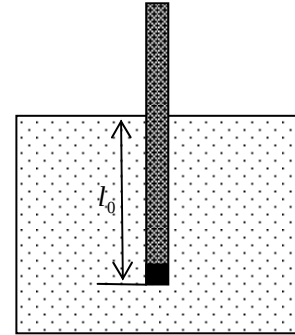
Substituting the result of t (b) into Eq. (3), we have

$$F_f = mg \sin \theta_0 + m\omega^2 d = mg \sin \theta_0 \left(1 + \frac{2md^2}{I_A}\right) = mg \sin \theta_0 \frac{I_A + 2md^2}{I_A}. \quad (5)$$

Therefore,

$$\tan \theta_0 = \frac{\mu I_C}{I_A + 2md^2} \quad \text{or} \quad \theta_0 = \arctan \frac{\mu I_C}{I_A + 2md^2} = \arctan \frac{\mu I_C}{I_C + 3md^2}.$$

(6) (20pts) A cylindrical wooden stick with lead attached at on end floats upright in water as shown in the figure. The total mass of the stick is m . In equilibrium, the length of submerged portion of the stick in water is l_0 . The density of the water is ρ and the area of the cross section of the stick is A .



(a) Write the equation of motion of the stick.

(b) Show that the oscillation of the stick is simple harmonic.

(c) Find the period of the oscillation.

Solution: (a) Let x be the deviation of the length of the submerged part of the stick to its equilibrium length l_0 . The equation of motion can be written as

$$m\ddot{x} = mg - \rho l A g.$$

Noting that $mg = \rho l_0 A g$, the equation can be rewritten as

$$m\ddot{x} = \rho l_0 A g - \rho l A g = -\rho A g x$$

or

$$\ddot{x} + \omega^2 x = 0, \quad \omega^2 = \frac{\rho A g}{m}.$$

(b) One can easily see from above result that the oscillation is simple harmonic.

(c) The circular frequency is

$$\omega = \sqrt{\frac{\rho A g}{m}}$$

and then the period is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{\rho A g}}.$$

(For reference only.)